Churn Statistical Analysis

Ruben Cabrera

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Load Libraries

```
required_pkgs <- c("readxl", "dplyr", "ggplot2", "caret", "randomForest", "broom")
new_pkgs <- required_pkgs[!required_pkgs %in% installed.packages()[, "Package"]]
if (length(new_pkgs)) install.packages(new_pkgs, repos = "https://cloud.r-project.org")

library(readxl)
library(dplyr)
library(ggplot2)
library(caret)
library(randomForest)
library(broom)
```

Load & Merge Data

Both files share the `Phone` and `Area Code` columns, which are used as join keys.

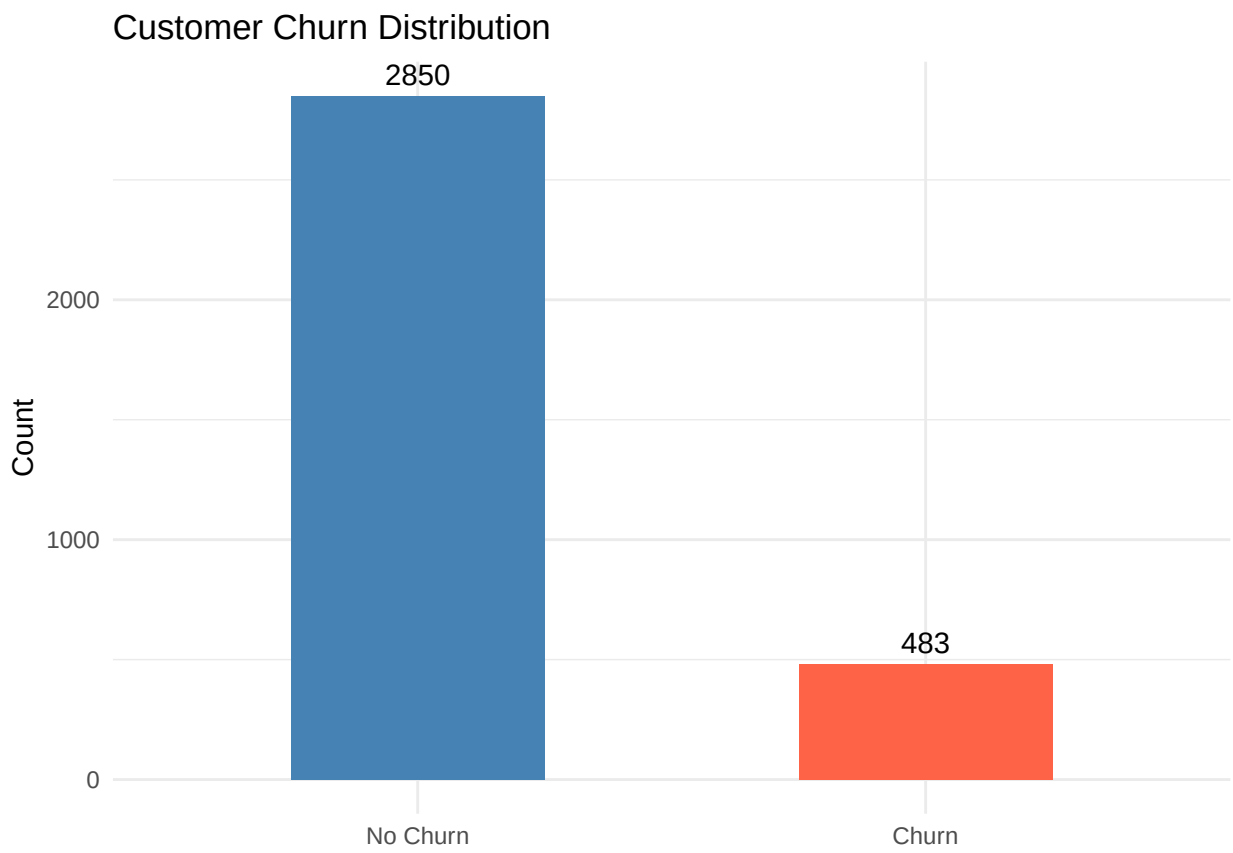
```
calls    <- read_xls("CallsData.xls")
contract <- read.csv("ContractData.csv", stringsAsFactors = TRUE)

# Standardise key column names before joining
# contract already reads "Area Code" as Area.Code; align calls to match
calls <- calls %>% rename(Area.Code = `Area Code`)

churn <- inner_join(contract, calls, by = c("Phone", "Area.Code"))

# Sanitise all column names (replace spaces/special chars with dots)
# so that formula-based models (glm, randomForest) work without issues
names(churn) <- make.names(names(churn), unique = TRUE)

# Convert binary columns to factors for cleaner plots
churn <- churn %>%
  mutate(
    Churn       = factor(Churn,       labels = c("No Churn", "Churn")),
    Int.l.Plan  = factor(Int.l.Plan,  labels = c("No Intl Plan", "Intl Plan")),
    VMail.Plan  = factor(VMail.Plan,  labels = c("No VMail", "VMail"))
  )

glimpse(churn)
```

```
## Rows: 3,333
## Columns: 21
## $ Account.Length <int> 128, 107, 137, 84, 75, 118, 121, 147, 117, 141, 65, 74,~
```

```
## $ Churn          <fct> No Churn, No Churn, No Churn, No Churn, No Churn, No Ch~
## $ Int.l.Plan     <fct> No Intl Plan, No Intl Plan, No Intl Plan, Intl Plan, In~
## $ VMail.Plan     <fct> VMail, VMail, No VMail, No VMail, No VMail, No VMail, V~
## $ State          <fct> KS, OH, NJ, OH, OK, AL, MA, MO, LA, WV, IN, RI, IA, MT,~
## $ Area.Code      <dbl> 415, 415, 415, 408, 415, 510, 510, 415, 408, 415, 415, ~
## $ Phone          <chr> "382-4657", "371-7191", "358-1921", "375-9999", "330-66~
## $ VMail.Message  <dbl> 25, 26, 0, 0, 0, 0, 24, 0, 0, 37, 0, 0, 0, 0, 0, 0, 27,~
## $ Day.Mins       <dbl> 265.1, 161.6, 243.4, 299.4, 166.7, 223.4, 218.2, 157.0,~
## $ Eve.Mins       <dbl> 197.4, 195.5, 121.2, 61.9, 148.3, 220.6, 348.5, 103.1, ~
## $ Night.Mins     <dbl> 244.7, 254.4, 162.6, 196.9, 186.9, 203.9, 212.6, 211.8,~
## $ Intl.Mins      <dbl> 10.0, 13.7, 12.2, 6.6, 10.1, 6.3, 7.5, 7.1, 8.7, 11.2, ~
## $ CustServ.Calls <dbl> 1, 1, 0, 2, 3, 0, 3, 0, 1, 0, 4, 0, 1, 3, 4, 4, 1, 3, 1~
## $ Day.Calls      <dbl> 110, 123, 114, 71, 113, 98, 88, 79, 97, 84, 137, 127, 9~
## $ Day.Charge     <dbl> 45.07, 27.47, 41.38, 50.90, 28.34, 37.98, 37.09, 26.69,~
## $ Eve.Calls      <dbl> 99, 103, 110, 88, 122, 101, 108, 94, 80, 111, 83, 148, ~
## $ Eve.Charge     <dbl> 16.78, 16.62, 10.30, 5.26, 12.61, 18.75, 29.62, 8.76, 2~
## $ Night.Calls    <dbl> 91, 103, 104, 89, 121, 118, 118, 96, 90, 97, 111, 94, 1~
## $ Night.Charge   <dbl> 11.01, 11.45, 7.32, 8.86, 8.41, 9.18, 9.57, 9.53, 9.71,~
## $ Intl.Calls     <dbl> 3, 3, 5, 7, 3, 6, 7, 6, 4, 5, 6, 5, 2, 5, 6, 9, 4, 3, 5~
## $ Intl.Charge    <dbl> 2.70, 3.70, 3.29, 1.78, 2.73, 1.70, 2.03, 1.92, 2.35, 3~
```

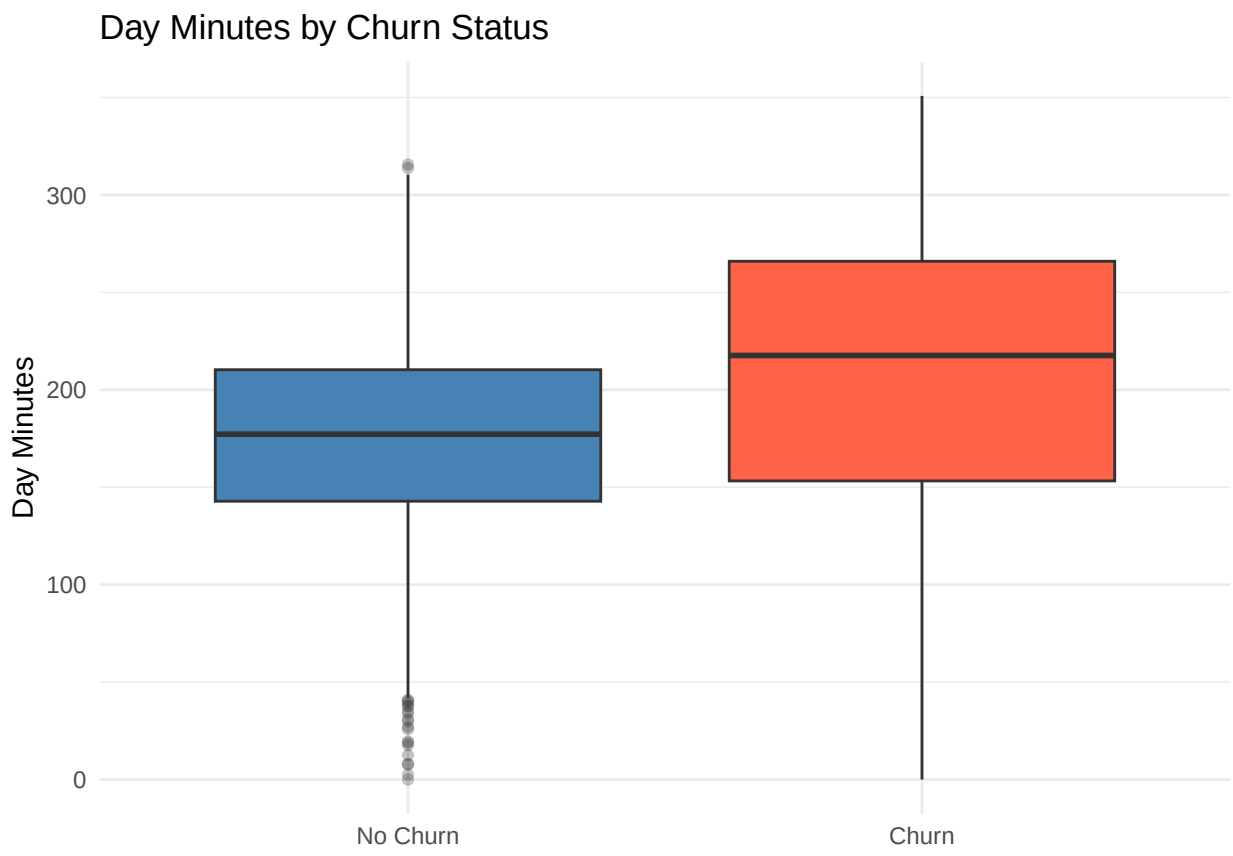
Plot 1 — Churn Distribution

```r
ggplot(churn, aes(x = Churn, fill = Churn)) +
  geom_bar(width = 0.5, show.legend = FALSE) +
  geom_text(stat = "count", aes(label = after_stat(count)), vjust = -0.5) +
  scale_fill_manual(values = c("steelblue", "tomato")) +
  labs(
    title = "Customer Churn Distribution",
    x     = NULL,
    y     = "Count"
  ) +
  theme_minimal()
```



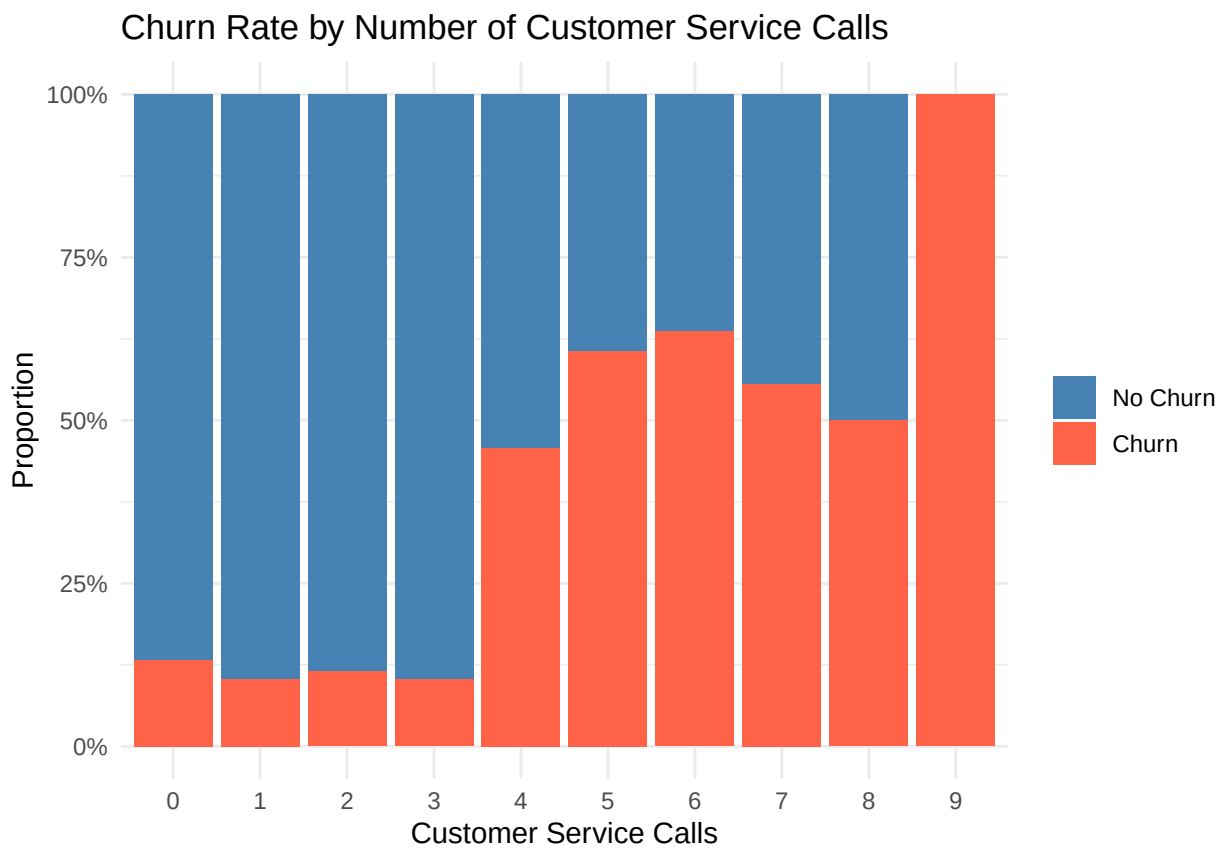
Plot 2 — Day Minutes by Churn

```
ggplot(churn, aes(x = Churn, y = Day.Mins, fill = Churn)) +
  geom_boxplot(show.legend = FALSE, outlier.alpha = 0.3) +
  scale_fill_manual(values = c("steelblue", "tomato")) +
  labs(
    title = "Day Minutes by Churn Status",
    x     = NULL,
    y     = "Day Minutes"
  ) +
  theme_minimal()
```



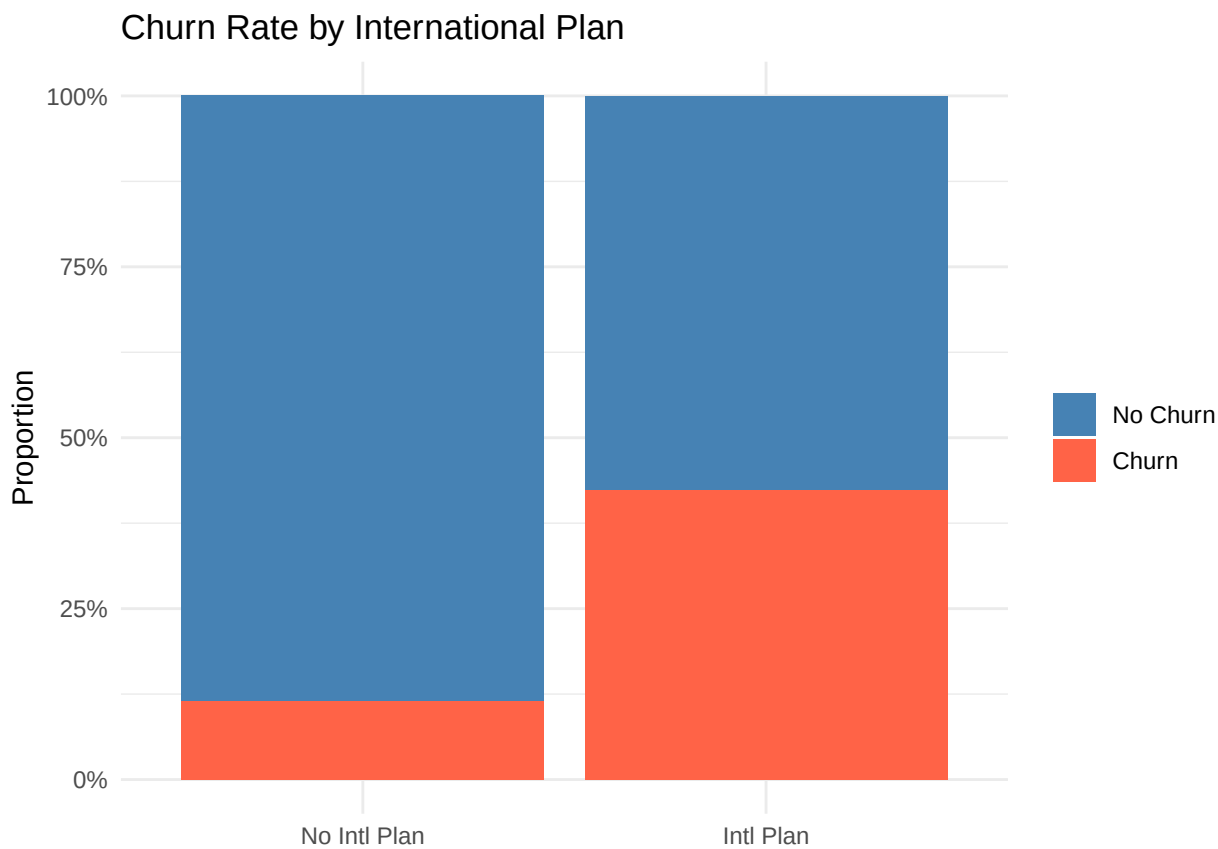
Plot 3 — Customer Service Calls by Churn

```
ggplot(churn, aes(x = factor(CustServ.Calls), fill = Churn)) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = scales::percent) +
  scale_fill_manual(values = c("steelblue", "tomato")) +
  labs(
    title = "Churn Rate by Number of Customer Service Calls",
    x     = "Customer Service Calls",
    y     = "Proportion",
    fill  = NULL
  ) +
  theme_minimal()
```



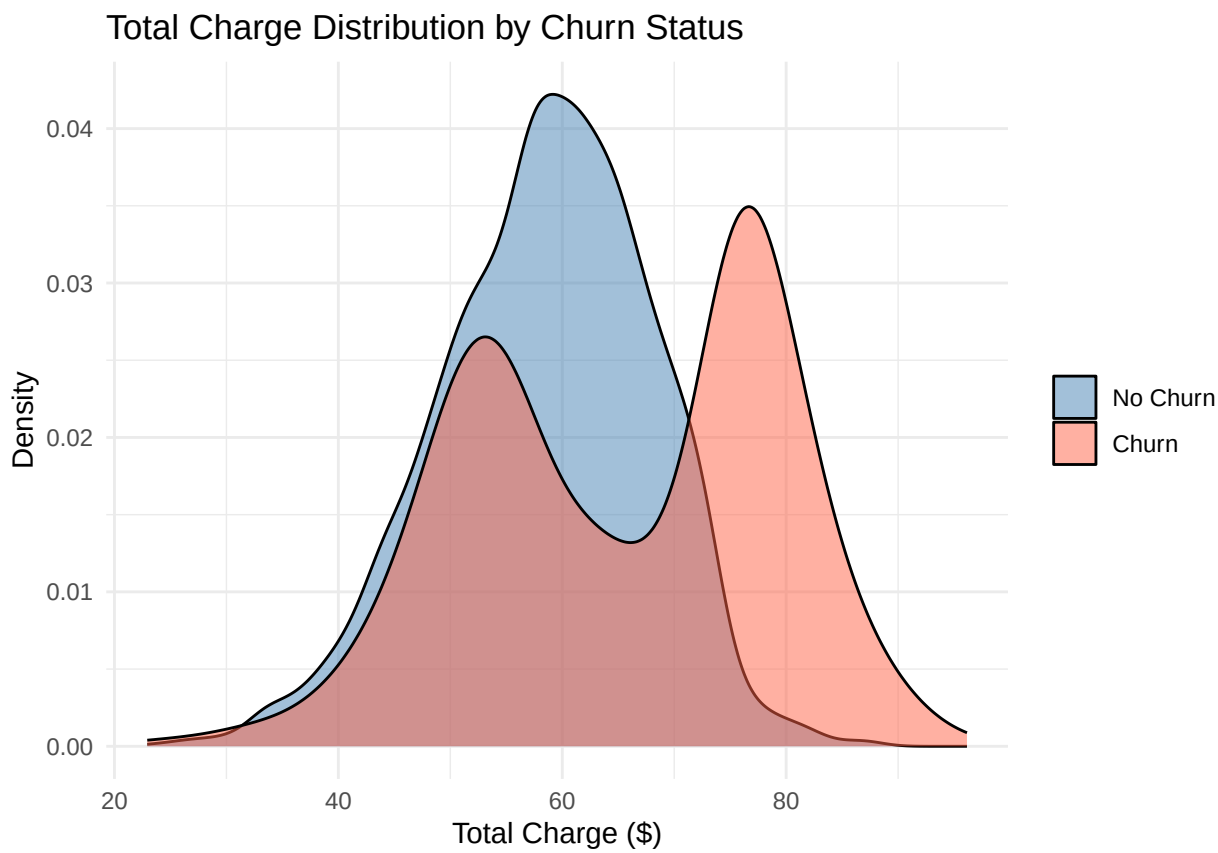
Plot 4 — International Plan vs Churn

```
ggplot(churn, aes(x = Int.l.Plan, fill = Churn)) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = scales::percent) +
  scale_fill_manual(values = c("steelblue", "tomato")) +
  labs(
    title = "Churn Rate by International Plan",
    x     = NULL,
    y     = "Proportion",
    fill  = NULL
  ) +
  theme_minimal()
```



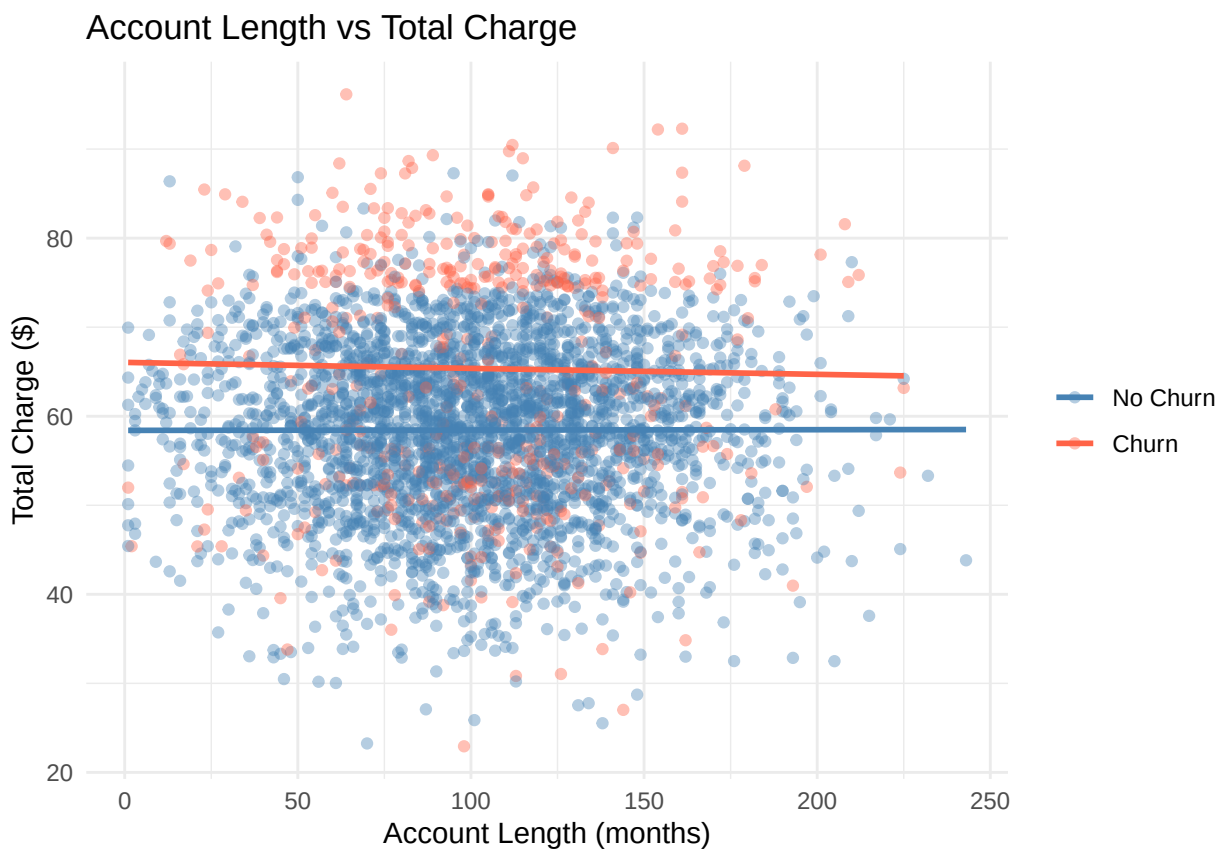
Plot 5 — Total Charges Distribution by Churn

```
churn <- churn %>%
  mutate(Total.Charge = Day.Charge + Eve.Charge + Night.Charge + Intl.Charge)

ggplot(churn, aes(x = Total.Charge, fill = Churn)) +
  geom_density(alpha = 0.5) +
  scale_fill_manual(values = c("steelblue", "tomato")) +
  labs(
    title = "Total Charge Distribution by Churn Status",
    x     = "Total Charge ($)",
    y     = "Density",
    fill  = NULL
  ) +
  theme_minimal()
```



Plot 6 — Account Length vs Total Charge

```
ggplot(churn, aes(x = Account.Length, y = Total.Charge, colour = Churn)) +
  geom_point(alpha = 0.4, size = 1.5) +
  geom_smooth(method = "lm", se = FALSE, linewidth = 1) +
  scale_colour_manual(values = c("steelblue", "tomato")) +
  labs(
    title  = "Account Length vs Total Charge",
    x      = "Account Length (months)",
    y      = "Total Charge ($)",
    colour = NULL
  ) +
  theme_minimal()
```



Predictive Modelling: Estimating Churn

The target variable `Churn` is binary (No Churn / Churn), so we need a **classification** model. Two well-suited candidates for this problem are:

Option 1 — Logistic Regression

Logistic Regression models the **probability** that a customer churns as a function of the predictors, using the logistic (sigmoid) function to keep outputs in $[0, 1]$:

$$P(\text{Churn} = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p)}}$$

Advantages:

- Highly interpretable — coefficients express the log-odds change per unit predictor increase.
- Fast to train, works well even with relatively small datasets.
- Provides calibrated probability estimates out of the box.

Limitations:

- Assumes a linear relationship between predictors and the log-odds of the outcome.
- Sensitive to highly correlated predictors (multicollinearity).

Option 2 — Random Forest

Random Forest builds an **ensemble of decision trees**, each trained on a random bootstrap sample of the data and a random subset of features. The final prediction is the majority vote across all trees.

Advantages:

- Captures non-linear relationships and interaction effects automatically.
- Robust to outliers and does not require feature scaling.
- Provides variable importance scores, helping identify the key drivers of churn.

Limitations:

- Less interpretable than Logistic Regression ("black-box" nature).
- More computationally expensive to train and tune.

Model Preparation

We proceed with **Logistic Regression** as the primary model given its interpretability, and use **Random Forest** as a complementary model to capture non-linear patterns. Both are evaluated on a held-out test set (30 % of the data).

```r
library(caret)

# Re-encode Churn as a clean binary factor (required by caret)
churn_model <- churn %>%
  mutate(Churn = factor(Churn, levels = c("No Churn", "Churn")))

# Drop identifier columns not useful for prediction
churn_model <- churn_model %>%
  select(-Phone, -State, -Area.Code)

# Train / test split (70 / 30)
set.seed(42)
train_idx <- createDataPartition(churn_model$Churn, p = 0.70, list = FALSE)
train_set <- churn_model[ train_idx, ]
test_set  <- churn_model[-train_idx, ]

cat("Train rows:", nrow(train_set), "| Test rows:", nrow(test_set), "\n")
```

```
## Train rows: 2334 | Test rows: 999
```

Model 1 — Logistic Regression

```r
log_model <- glm(Churn ~ ., data = train_set, family = binomial)

summary(log_model)
```

```
##
## Call:
## glm(formula = Churn ~ ., family = binomial, data = train_set)
##
## Coefficients: (1 not defined because of singularities)
##                        Estimate Std. Error z value Pr(>|z|)
```

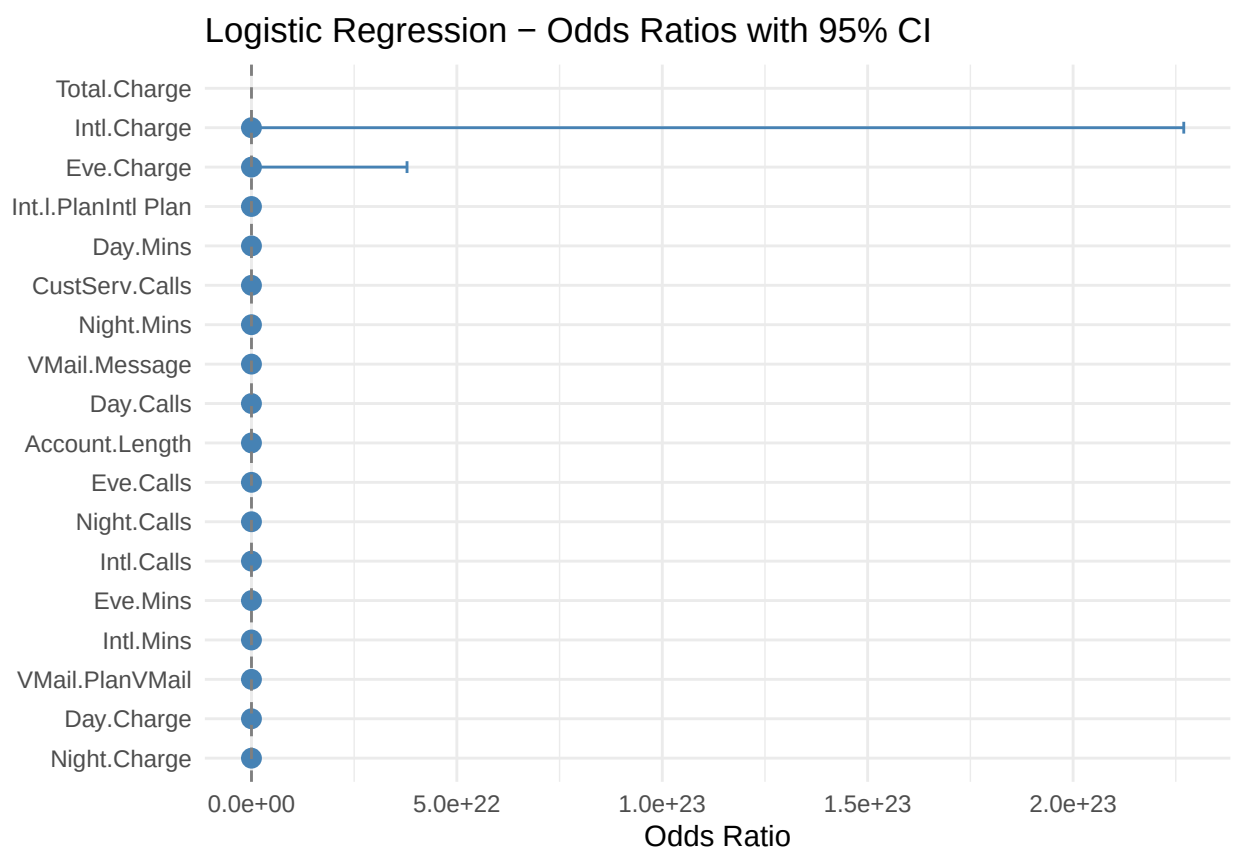
```
## (Intercept)         -8.3926875  0.8519684  -9.851  < 2e-16 ***
## Account.Length       0.0009814  0.0016345   0.600 0.548201
## Int.l.PlanIntl Plan  1.9542851  0.1763589  11.081  < 2e-16 ***
## VMail.PlanVMail     -2.4804379  0.7312631  -3.392 0.000694 ***
## VMail.Message        0.0458534  0.0226172   2.027 0.042625 *
## Day.Mins             0.8738528  3.9176316   0.223 0.823492
## Eve.Mins            -0.5721088  1.9576214  -0.292 0.770098
## Night.Mins           0.4448984  1.0496257   0.424 0.671665
## Intl.Mins           -1.8973984  6.3981776  -0.297 0.766808
## CustServ.Calls       0.5099597  0.0474170  10.755  < 2e-16 ***
## Day.Calls            0.0052843  0.0032699   1.616 0.106081
## Day.Charge          -5.0592092 23.0447236  -0.220 0.826230
## Eve.Calls           -0.0026245  0.0033677  -0.779 0.435799
## Eve.Charge           6.8237807 23.0307091   0.296 0.767008
## Night.Calls         -0.0028942  0.0034092  -0.849 0.395907
## Night.Charge        -9.7813943 23.3239208  -0.419 0.674944
## Intl.Calls          -0.0964826  0.0298614  -3.231 0.001234 **
## Intl.Charge          7.2783319 23.6948053   0.307 0.758714
## Total.Charge                NA         NA      NA       NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1934.3  on 2333  degrees of freedom
## Residual deviance: 1506.8  on 2316  degrees of freedom
## AIC: 1542.8
##
## Number of Fisher Scoring iterations: 6
```

```r
log_probs <- predict(log_model, newdata = test_set, type = "response")
log_preds <- factor(ifelse(log_probs >= 0.5, "Churn", "No Churn"),
                    levels = c("No Churn", "Churn"))

cm_log <- confusionMatrix(log_preds, test_set$Churn, positive = "Churn")
print(cm_log)
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction No Churn Churn
##   No Churn      831   114
##   Churn          24    30
##
##                Accuracy : 0.8619
##                  95% CI : (0.8389, 0.8827)
##     No Information Rate : 0.8559
##     P-Value [Acc > NIR] : 0.313
##
##                   Kappa : 0.2436
##
##  Mcnemar's Test P-Value : 3.559e-14
##
##             Sensitivity : 0.20833
##             Specificity : 0.97193
```

```
##          Pos Pred Value : 0.55556
##          Neg Pred Value : 0.87937
##              Prevalence : 0.14414
##          Detection Rate : 0.03003
##    Detection Prevalence : 0.05405
##       Balanced Accuracy : 0.59013
##
##        'Positive' Class : Churn
##
```

```r
# Visualise significant predictors via odds ratios
log_coefs <- broom::tidy(log_model, exponentiate = TRUE, conf.int = TRUE) %>%
  filter(term != "(Intercept)") %>%
  arrange(desc(estimate))

ggplot(log_coefs, aes(x = reorder(term, estimate), y = estimate)) +
  geom_point(size = 3, colour = "steelblue") +
  geom_errorbar(aes(ymin = conf.low, ymax = conf.high), width = 0.3, colour = "steelblue") +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey50") +
  coord_flip() +
  labs(
    title = "Logistic Regression - Odds Ratios with 95% CI",
    x     = NULL,
    y     = "Odds Ratio"
  ) +
  theme_minimal()
```



Model 2 — Random Forest

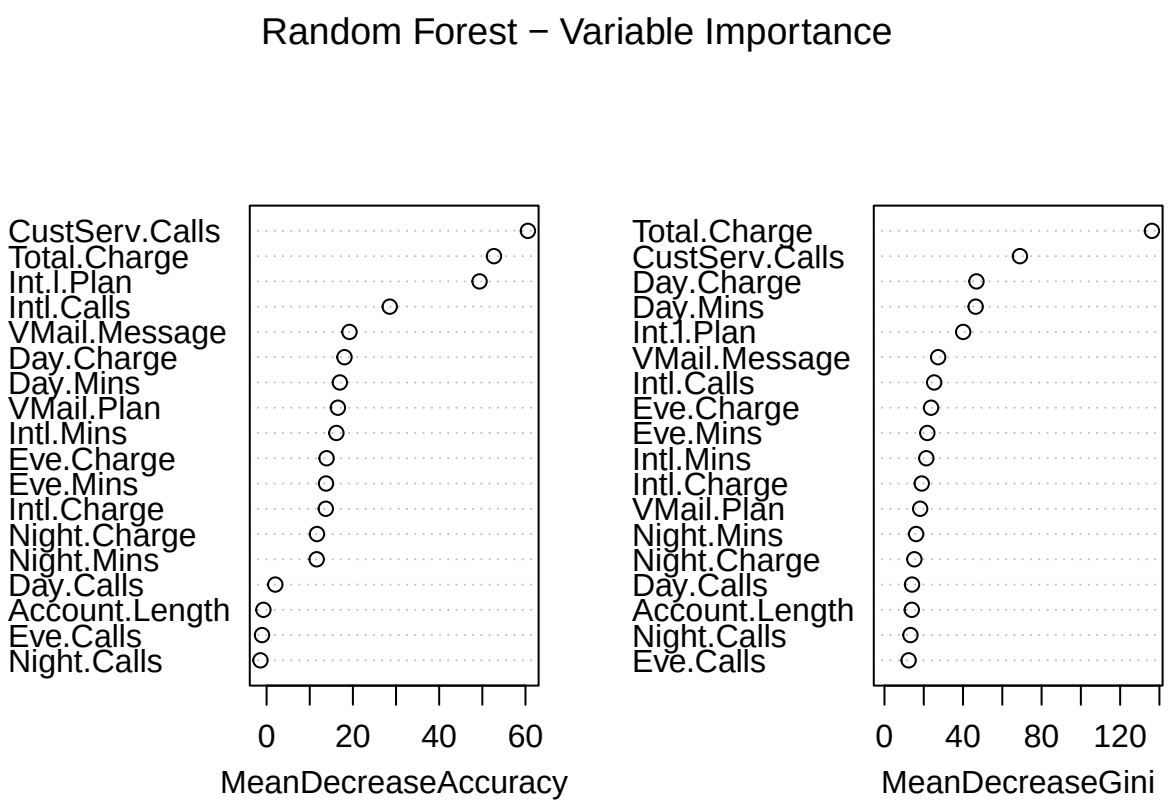
```r
library(randomForest)

set.seed(42)
rf_model <- randomForest(Churn ~ ., data = train_set, ntree = 300, importance = TRUE)

print(rf_model)
```

```
##
## Call:
##  randomForest(formula = Churn ~ ., data = train_set, ntree = 300,      importance = TRUE)
##                Type of random forest: classification
##                      Number of trees: 300
## No. of variables tried at each split: 4
##
##         OOB estimate of  error rate: 2.4%
## Confusion matrix:
##          No Churn Churn  class.error
## No Churn     1994     1 0.0005012531
## Churn          55   284 0.1622418879
```

```r
rf_preds <- predict(rf_model, newdata = test_set)

cm_rf <- confusionMatrix(rf_preds, test_set$Churn, positive = "Churn")
print(cm_rf)
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction No Churn Churn
##   No Churn      855    20
##   Churn           0   124
##
##                Accuracy : 0.98
##                  95% CI : (0.9692, 0.9877)
##     No Information Rate : 0.8559
##     P-Value [Acc > NIR] : < 2.2e-16
##
##                   Kappa : 0.9139
##
##  Mcnemar's Test P-Value : 2.152e-05
##
##             Sensitivity : 0.8611
##             Specificity : 1.0000
##          Pos Pred Value : 1.0000
##          Neg Pred Value : 0.9771
##              Prevalence : 0.1441
##          Detection Rate : 0.1241
##    Detection Prevalence : 0.1241
##       Balanced Accuracy : 0.9306
##
##        'Positive' Class : Churn
```

```
##
# Variable importance plot
varImpPlot(rf_model, main = "Random Forest - Variable Importance")
```



Model Comparison

```
results <- data.frame(
  Model     = c("Logistic Regression", "Random Forest"),
  Accuracy  = c(cm_log$overall["Accuracy"],  cm_rf$overall["Accuracy"]),
  Precision = c(cm_log$byClass["Precision"], cm_rf$byClass["Precision"]),
  Recall    = c(cm_log$byClass["Recall"],    cm_rf$byClass["Recall"]),
  F1        = c(cm_log$byClass["F1"],        cm_rf$byClass["F1"])
)

knitr::kable(results, digits = 4, caption = "Test-set performance comparison")
```

Table 1: Test-set performance comparison

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.8619	0.5556	0.2083	0.3030
Random Forest	0.9800	1.0000	0.8611	0.9254

Conclusions

Overall Performance

Random Forest clearly outperforms Logistic Regression across every metric on the held-out test set (30 % of 3,333 observations):

Metric	Logistic Regression	Random Forest
Accuracy	86.19 %	**98.00 %**
Precision	55.56 %	**100.00 %**
Recall	20.83 %	**86.11 %**
F1	0.303	**0.925**

Logistic Regression

Although it achieves a seemingly acceptable accuracy of **86.2 %**, this figure is misleading given the class imbalance in the dataset (~14 % churners). The model struggles to identify actual churners: its **Recall of only 20.8 %** means it misses roughly 4 out of every 5 customers who will churn — a critical failure for a retention use case where false negatives are costly. Its low F1 score (0.30) confirms that it is not a reliable churn detector.

Random Forest

Random Forest achieves **98 % accuracy** with a **Precision of 100 %** (no false alarms) and a **Recall of 86.1 %** (detects the vast majority of true churners), yielding an F1 of **0.93**. This makes it the clearly preferred model for deployment in a churn-prevention campaign.

Key Drivers of Churn (Random Forest Variable Importance)

The top predictors ranked by Mean Decrease in Gini impurity are:

1. **Total Charge** (136.17) — the single strongest predictor; heavy spenders are more likely to leave.
2. **Customer Service Calls** (68.97) — repeated contacts signal dissatisfaction and strongly anticipate churn.
3. **Day Charge / Day Mins** (~46) — high daytime usage costs drive customers away.
4. **International Plan** (40.08) — customers on the international plan churn at a disproportionately high rate.
5. **VMail Messages** (27.33) — lower voicemail engagement is associated with churn.

Recommendations

- **Prioritise high-billing customers** for proactive retention offers before they reach the churn threshold.
- **Flag customers with 3 service calls** for immediate intervention; dissatisfaction escalates quickly.
- **Review the International Plan pricing** — its strong association with churn suggests it may not deliver perceived value.
- For production use, consider **threshold tuning** on the Random Forest probability output to further optimise the Recall/Precision trade-off according to the cost of false negatives vs. false positives.